

bridge_phase2_p1_r1_review_2026061

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Independent Code Review — Bridge Phase-2 No-Local-Runtime Redirect (P-1 → R-1)

Revision: 1 **Last modified:** 2026-06-15T21:01:10Z **Reviewer:** independent code-review agent (structurally separate from the author, §11.4.142 / §11.4.125)
Authority: §11.4.142 (every change reviewed), §11.4.125 (review gate before build), §11.4.134 (iterate-until-GO), §11.4.69 (no silent fallback) **Scope:** parent repo /Volumes/T7/Projects/helix_translate, commits e5307ce..35cb12c reviewed as one logical change.

Verdict: GO (no BLOCKING findings)

The no-local-runtime redirect is correct, the R2 no-silent-fallback mandate holds on every redirected path, the new tests genuinely catch mis-wiring, the nil-default seams are behaviour-preserving, the deferrals (R-3/R-4) are correctly scoped, and the 3 reported failures are git-proven pre-existing and unrelated. **R-2/R-3 removals may proceed.** Three NIT-level hardening items are listed (none gate the build).

Build green at HEAD (go build ./cmd/... ./pkg/... → BUILD_OK). go vet clean on all touched packages. All redirect test suites pass at HEAD:

ok	pkg/bridge	ok	pkg/coordination	ok	pkg/preparation
ok	pkg/verification				
ok	pkg/api (Bridge)	ok	pkg/grpc (Bridge)	ok	cmd/unified-translator (Bridge)

Commits reviewed

- P-1 e5307ce — bridge.BestClient adapter + Invoke extract-method refactor
- P-1.5 86816a4 — ProviderDiverse* API + injectable EnsembleTranslatorFactory seams

- R-1 9f6be23 / 0713bec / 3cd4a77 / 0a3b607 / 7f843fc / 15cab94 / 35cb12c — adapters + ~10 single-site + 3 ensemble-site redirects + R2 honest no-key error
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Dimension 1 — Correctness — GO

- **P-1 Invoke refactor is behaviour-preserving (FACT).** `git show e5307ce` shows a pure extract-method: `resolveBest` (bridge.go:382) is the verbatim prior inline selection→resolution→marker logic; `realClientBuild` (bridge.go:313) is the verbatim prior `NewOpenAIClient` call. `Invoke` (bridge.go:425) and the new `BestClient` (bridge.go:413) both route through the single `clientBuild` construction path — exactly ONE construction path, as documented.
- **Bridge opened once per binary, not per-request.** `apiHandler.bridgeFor` (handler.go:~790) and `grpcCoreTranslatorImpl.bridgeFor` (core_translator.go) both lazy-open via `sync.Once` (bridgeOnce.Do) caching `bridge+bridgeErr`; `cli/server/markdown/preparation` mains `bridge.Open(...)` once in main. No per-request leak.
- **sync.Once lazy handlers are thread-safe.** `bridge/bridgeErr` are written only inside `Once.Do` and read only after `Do` returns — the standard safe `sync.Once` memo pattern. No data race.
- **Provider-diverse adapter genuinely gives one-per-provider, no collapse (FACT).** `ProviderDiverseModels` (bridge.go:455-480) dedups via a seen `map[providerID]bool` over the score-descending `ListVerified`, keeping first-per-provider (= strongest per provider). `FallbackOrder` re-numbered 1..N. `ProviderDiverseClients/Translators` build through the same whitelist-immune `clientBuild` seam. Mutation-proven by `TestBridge_ProviderDiverseClients_OnePerProviderRankedByScore` (RED FAILs got 3, want 5).
- **clientTranslator.Translate threads contextStr verbatim** (bridge.go:554-555) — no argument loss. Sole ensemble consumer is `polisher.go:563 translator.Translate(ctx, prompt, location)`, matching the thin surface (FACT).

Dimension 2 — R2 no-silent-fallback (core mandate) — GO

Every redirected default path returns an honest hard error on no-key and CANNOT reach a local runtime:

- **bridge.Open is the choke point** (bridge.go:98). No `*_API_KEY` → `bootstrapInProgress` returns the honest error "no provider API keys set ... (local llama.cpp fallback is not permitted)" (bridge.go:160-164). No code path constructs `llama.cpp/Ollama` from `Open`.
- **api createTranslator** (handler.go:790) routes everything except `provider=="distributed"` through `bridgeFor()`; the old `llm.NewLLMTranslator(config)` is deleted. `bridgeErr` is cached and surfaced, never masked.
- **grpc executeTranslation switch** (core_translator.go:255) — `ssh`→SSH (R-3 deferred), `llamacpp`→explicit operator-chosen local (R-3 deferred),

- default**→**executeAPITranslation**→**bridge**. The deleted `llm.NewLLMTranslator` is replaced by `ct.bridgeFor()`. Local `llama.cpp` reachable ONLY by explicit `ProviderConfig.Type=="llamacpp"` (operator choice, not silent fallback).
- **cli** (main.go) — bridge opened once; multi-LLM via `NewMultiLLMTranslatorWrapperWithFactory(..., b.EnsembleFactory(task)), single fallback via b.BestTranslatorFunc(task)(ctx); no-key → return fmt.Errorf("LLMsVerifier bridge unavailable (no local-runtime fallback): %w", ...).`
- **server** (main.go) — `NewMultiLLMCoordinatorWithFactory(..., b.EnsembleFactory(...))`; no-key → `log.Fatalf("LLMsVerifier bridge unavailable (no local-runtime fallback)")`.
- **markdown-translator** (main.go) — the old `createTranslator` case "llamacpp" no-key local arm is DELETED; replaced by `createTranslator(ctx, b, task)→b.BestTranslatorFunc`. No-key → `log.Fatalf`.
- **preparation-translator** (main.go) — base translator now `b.BestTranslator(ctx, task)` instead of hardcoded `deepseek` `NewLLMTranslator`; no-key → fatal.
- **unified-translator** (main.go:488) — `executeAPITranslation` default branch → `bridgeTranslator→bridge.Open+BestTranslator`; mock is an explicit offline seam, `ssh/llamacpp` arms untouched (R-3/R-4). No-key → "bridge translation unavailable (no local-runtime fallback)".

The remaining `llamacpp/ollama` constructors in the tree (`cmd/unified-translator:307/448`, `pkg/grpc/core_translator.go:259`, `pkg/translator/llm/llamacpp*.go`, `cmd/translate-ssh`) are all reachable ONLY via an EXPLICIT operator provider selection (R-3/R-4 deferred scope) — never from a redirected default. **No silent-fallback path found.**

Dimension 3 — Test genuineness (§1.1 / §11.4.1) — GO

Independent adversarial sub-review (full transcript on file) examined all redirect tests, ran each offline, and verified all six `RED_MODE` polarities genuinely FAIL on the broken branch. Findings:

- **pkg/bridge/bridge_test.go** — **GENUINE**. `invokeDispatch/clientBuild` are real package vars on the single production routing path; guards independently recompute the oracle via `BestModel` and assert captured (`providerMarker, modelID`) equals it. `TestBridge_Invoke_RoutesToBestModel` RED proven FAIL (routed to "deepseek-chat", want "gpt-4o"). The `BestClient` provider-name pair deliberately uses a **deepseek** strongest model so the bare `*OpenAIClient` "openai" would FAIL — load-bearing against the exact hardcoded-openai defect. `TestOpen_NoKeys_HonestHardError` deterministic via empty `Getenv`, asserts the R2 error text + that any "llama" mention is framed "not permitted".
- **pkg/api / pkg/grpc bridge_redirect_test.go** — **GENUINE**. Inject a fake `bridgeTranslatorFactory`, assert the fake's `BRIDGE:/GetName()=="bridge-verified"` output is actually returned (not the old `NewLLMTranslator` path) AND lang pair threaded. No-key tests run the REAL

bridge.Open with empty Getenv → assert "no provider API keys".

Distributed-arm test installs a factory that t.Fatals if reached.

- **cmd/unified-translator/bridge_redirect_test.go** — **GENUINE**. The mock-seam IS load-bearing: removing case "mock" drops it to the bridge default → real os.Getenv → mis-route FAILs the err==nil assertion.
- **coordination/preparation/verification *_factory_test.go** — **GENUINE**. RED polarity (factory not injected / set on a discarded struct) proven FAIL on all three; GREEN asserts the factory is reached with the run's ctx (sentinel) and yields the exact injected provider instances.

No tautology that would pass on a wrong route was found.

Dimension 4 — nil-default seam behaviour-preservation — GO

- MultiLLMCoordinator.translatorFactory is consumed at exactly ONE branch — initializeLLMInstances line 141
if c.translatorFactory != nil { initializeFromFactory();
return }. When nil (un-redirected callers, config.TranslatorFactory zero value), control falls through to the unchanged discoverProviders + NewLLMTranslator loop. Docstring claim "byte-for-byte identical to before" is FACT-accurate.
- Original NewMultiLLMCoordinator (line 97) retained with unchanged signature; new NewMultiLLMCoordinatorWithFactory (line 129) is additive. The same additive nil-default pattern applies to preparation/verification per the P-1.5 + R-1c diffs and their _NilFactoryUnchanged companion tests.

Dimension 5 — regression blast-radius — GO

- Changed signatures (createTranslator, executeAPITranslation, the *WithFactory constructors) verified: callers compile (build green) and the original no-factory constructors are preserved, so un-redirected callers are unbroken.
- **The 3 reported failures are git-proven PRE-EXISTING and unrelated** (independent sub-review, full transcript on file). git diff --stat 49d94f2..35cb12c is EMPTY for internal/verifier/, test/distributed/, pkg/distributed/, and the SSH test — the redirect touched none of them, so it definitionally cannot have caused them.
 - **TestAffirmativeResponseIsHardGate** — internal/verifier/pipeline_affirmative_gate_test.go. The default suite PASSES (ok ... 0.194s); it FAILs ONLY under explicit RED_MODE=1 (by-design §11.4.115 reproduction). The GREEN-default polarity was set by 97a8afd, an ancestor of baseline 49d94f2. **CORRECTION to author's framing:** it is NOT "in default-RED mode" — default is GREEN/PASS. Not a regression; just imprecise wording in the handoff.
 - **test/distributed race** — manager_test.go:587–588 unguarded worker.ActiveJobs++ in the test's own mock; byte-identical at baseline; surfaces only under -race.
 - **test/integration SSH** — TestSSHTranslationIntegration PASSES at HEAD via an in-process mock SSH server; only the e2e variants topology-SKIP. Not a failure at HEAD.

Dimension 6 — deferred R-1d correctness — GO

- `modelsbridge.New` — **zero callers** in `pkg/cmd/internal/` (grep empty). No live consumer left broken; deferring to R-4 is correct.
- `markdown LLMProvider` (`llm.LLMClient` consumer) — the only non-test construction site is `cmd/translate-ssh/main.go:231` which sets it to `nil` ("Will be created remotely"). Non-`nil` only in the SSH-worker path (R-3/R-4 deferred). No consumer silently broken; deferral correct.

Dimension 7 — safety — GO

- No host/data regression: changes are in-process translator wiring only; no destructive ops, no power-state/host-session calls.
 - No secret leakage: `ModelInfo` is key-free (§11.4.10); keys read in-memory via `ProviderResolver/SetKeyResolver` and never returned/logged. `bridge_test.go` no-key tests use an empty `Getenv` override (no real keys touched).
 - Build green at HEAD; vet clean. History is sequential additive commits (P-1 prerequisite → P-1.5 seams → R-1c threading → per-binary redirects), each replacing a `NewLLMTranslator` site with a bridge call while keeping the original constructors — consistent with green-at-each-commit.
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NIT-level hardening (non-blocking — do NOT gate R-2/R-3)

1. **`bridge_test.go:436 TestBridge_ProviderDiverseTranslators...`** asserts only `len==3` + non-`nil`; add provider-identity + order assertions (or a `RED_MODE` polarity) to match its `...Clients` sibling. Currently covered transitively via the shared `ProviderDiverseModels` path, so not blocking.
 2. **`cmd/unified-translator/bridge_redirect_test.go:34`** — make the mock-seam guard *positively* assert the bridge was not reached (inject a `bridgeOpener` that `t.Fatals`, mirroring `api`'s distributed-arm test), and soften the env-dependent "live network attempt" docstring.
 3. **Docstring trims** — `TestBridge_NumericProviderID_HTTPPath` exercises resolution, not dispatch ("end-to-end at the bridge layer" overstates); and the handoff note for `TestAffirmativeResponseIsHardGate` should be corrected from "default-RED" to "GREEN-default; RED only under `RED_MODE=1`".
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§11.4.134 iterate-until-GO

No BLOCKING findings → the review loop terminates at a clean **GO** this pass. The 3 NITs are tracked here for the author but do not re-arm the loop (they introduce no defect and no false-result/bluff capability). The author may address them opportunistically; they are not preconditions for R-2/R-3 removals.